

DATA PLATFORM

The Problems We Solved — and Why We're Proud of It

A self-service data product built on CDC · Kinesis · ECS Fargate · MySQL system-of-record · Elasticsearch · S3 + Athena

Weeks → 1 Day

consumer onboarding

0

extra load on source DBs

Sub-second

search at scale

Nine Hard Problems — Solved by Design

Each one maps to a metric a client can put in front of leadership.



01

Freed our data without risking source systems



02

One source of truth, many shapes of truth



03

Producers and consumers fully decoupled



04

Guaranteed delivery — nothing gets lost



05

Fast search without breaking the database



06

Analysts self-serve — no more export tickets



07

Security & access solved once, centrally



08

Scales elastically — no re-architecting



09

Full observability — we sleep at night

01



Freed our data without risking source systems

METRIC THAT PROVES IT

Extra query load on source systems

BEFORE

Dozens of direct queries/sec hitting
production tables



AFTER

0 — CDC reads the binlog, never
the live tables



BUSINESS VALUE

**Zero consumer-driven outages on
systems-of-record.**

02



**One source of truth,
many shapes of
truth**

METRIC THAT PROVES IT

Data drift / sync incidents across systems

BEFORE

Frequent “search says X, API says Y”
tickets



AFTER

Near-zero — one spine fans out
consistently



BUSINESS VALUE

**Trust in data; reconciliation hours
reclaimed.**

03



Producers and consumers fully decoupled

METRIC THAT PROVES IT

Time & effort to onboard a new consumer

BEFORE

Weeks of fragile point-to-point integration



AFTER

~1 day, config-only — no upstream change



BUSINESS VALUE

Faster time-to-market; no cross-team coordination tax.

04



Guaranteed delivery — nothing gets lost

METRIC THAT PROVES IT

Message loss / unrecoverable events

BEFORE

Silent drops, discovered only by complaints



AFTER

0 lost — retry ×5 + DLQ + checkpoints



BUSINESS VALUE

A pipeline built for auditing, easy replay, and compliance.

05



Fast search without breaking the database

METRIC THAT PROVES IT

Search latency + DB CPU under load

BEFORE

Slow queries stressing the MySQL primary



AFTER

Sub-second search, primary DB untouched



BUSINESS VALUE

Better UX; headroom preserved on the core DB.

06



Analysts self-serve — no more export tickets

METRIC THAT PROVES IT

Engineering hours on data-extract
requests

BEFORE

Constant ad-hoc “please export data”
queue



AFTER

Near-zero — Athena / Spark on
the S3 lake



BUSINESS VALUE

**Engineers freed for real work;
analysts unblocked.**

07



**Security & access
solved once,
centrally**

METRIC THAT PROVES IT

Access-review time + auth consistency

BEFORE

Per-service auth; reviews taking days



AFTER

One JWT model — audited in
minutes



BUSINESS VALUE

**Consistent enforcement; faster
compliance sign-off.**

08



**Scales elastically —
no re-architecting**

METRIC THAT PROVES IT

Effort to absorb spikes & growth

BEFORE

Re-platforming proved expensive,
drawing on substantial CapEx



AFTER

Scale out tasks — same design



BUSINESS VALUE

**Growth absorbed without expensive
rebuilt.**

09



**Full observability —
we sleep at night**

METRIC THAT PROVES IT

Mean-Time-To-Detect (MTTD) incidents

BEFORE

Silent failures found by customers
first



AFTER

Caught proactively before
impact



BUSINESS VALUE

**Higher uptime; incidents resolved
before customers notice.**

OVERVIEW



*We turned locked-away transactional data into a safe, self-service data product — **search, APIs, real-time feeds and analytics** — with onboarding cut from **weeks to a day**, **zero extra load** on source systems, **zero data loss**, sub-second search and **10× scalability** — all without ever putting our core business at risk.*